

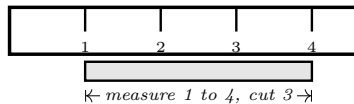
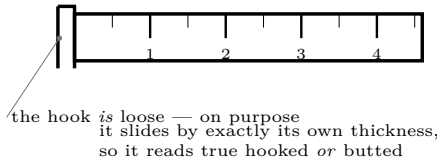
Craft 1: Measure & Mark

Everything downstream inherits the error you make here

Shop skill — transferable to anything you ever build

THE STANDARD

A mark is **one line**, thin enough that you can tell which side of it to cut on, drawn square around the whole circumference of round stock, and made from a measurement you took twice. If you cannot say which side of your own mark the material comes off, the mark is not good enough.



The burnt inch. Start at a whole number, not the hook. Subtract it. A worn hook then cannot lie to you.

Measure twice

Not a proverb — a procedure. Take the measurement, put the tape down, do something else for ten seconds, take it again. If the two disagree, take a third. Two agreeing measurements is evidence; one is a hope.

Mark thin

A pencil line is about half a millimetre wide, which on four joints is two millimetres of accumulated slop. Use a fine mechanical pencil, or scribe with a knife where a scratch is acceptable.

Mark which side

Put a small **X** on the waste side of every line, immediately. Cut so the blade eats the X and the line survives. This single habit removes most short-piece mistakes.

Go all the way round

On pipe, a mark on one side tells you nothing about the far side. Roll the pipe against a square, or wrap paper round it, and carry the line the whole way.

One reference Measure every cut from the *same* end of the stock, not from the last cut. Measuring cut-to-cut adds every previous error to the next one.

Write it down Cut list first, on paper, before any tool comes out. see the **Build Log sheet**

The wrap trick, and why it works

Wrap a sheet of paper around the pipe so the edge meets itself exactly. Because the paper edge is straight and the pipe is round, the line where the paper overlaps is automatically perpendicular to the pipe axis. Draw along it.

You should see: The line closes on itself when you have gone all the way round. If the two ends of the line miss each other, the paper was not aligned and the mark is not square — start again rather than splitting the difference.

Where error comes from, in order of size

Source	Typical	What to do
Measuring cut-to-cut	several mm	Always measure from one reference end.
Thick pencil line	0.5 mm each	Fine pencil or knife; cut to one consistent side.
Tape not flat or not straight	1–2 mm	Pull it taut against the work, sight down it.
Reading the wrong graduation	1.5 mm or more	Say the number out loud as you read it.
Worn or bent hook	up to 1 mm	Use the burnt inch and it cannot matter.
Parallax — eye off to one side	under 1 mm	Look straight down at the mark.

Check it yourself. Lay your marked pipe on the bench and roll it slowly. The line should stay at the same height all the way round — if it wanders up and down, it is not square and the cut made from it will not be either. Then measure the marked length again, from the reference end, with the tape you did not use the first time if you have two.

Why this sheet is first
Every later skill is downstream of this one. A cut cannot be squarer than its mark; a joint cannot be tighter than its cut; a chamber cannot be stronger than its joint. Accuracy is not a talent you have or lack — it is a set of small habits, and they are all on this page.

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